Additional Post-Hoc Noise Robustness Analysis
Hardware-class recall under sensor measurement noise

Noise	Coverage	Model	Short recall	Degradation recall	Open recall	HW macro recall
Clean	100.0000%	RF E4	0.9996	0.9907	1.0000	0.9968
Clean	100.0000%	LightGBM E4	0.4994	0.9930	0.9992	0.8305
1%	92.1941 ± 0.0737%	RF E4	0.9994 ± 0.0004	0.9895 ± 0.0012	0.9958 ± 0.0019	0.9949 ± 0.0007
1%	92.1941 ± 0.0737%	LightGBM E4	0.4971 ± 0.0011	0.9928 ± 0.0007	0.9954 ± 0.0018	0.8284 ± 0.0007
2%	84.4133 ± 0.0997%	RF E4	0.9979 ± 0.0007	0.9835 ± 0.0010	0.9472 ± 0.0036	0.9762 ± 0.0013
2%	84.4133 ± 0.0997%	LightGBM E4	0.4903 ± 0.0033	0.9870 ± 0.0021	0.9467 ± 0.0035	0.8080 ± 0.0019
5%	74.7809 ± 0.1046%	RF E4	0.9812 ± 0.0032	0.8869 ± 0.0073	0.8120 ± 0.0077	0.8934 ± 0.0029
5%	74.7809 ± 0.1046%	LightGBM E4	0.4288 ± 0.0043	0.8171 ± 0.0067	0.8096 ± 0.0075	0.6852 ± 0.0027
10%	69.5123 ± 0.1124%	RF E4	0.8792 ± 0.0067	0.6584 ± 0.0076	0.7198 ± 0.0073	0.7525 ± 0.0039
10%	69.5123 ± 0.1124%	LightGBM E4	0.3377 ± 0.0089	0.5643 ± 0.0099	0.7089 ± 0.0066	0.5370 ± 0.0036

Coverage is identical for RF and LightGBM; one shared column. Noisy values: mean ± sample SD across 10 paired seeds.

Coverage decreases because the unchanged input-quality rules can convert noise-induced negative electrical readings to missing values and trigger abstention.

End-to-end pipeline robustness under Gaussian sensor noise, including existing input-quality/abstention rules.